Cover Letter Cum Research Background

March 26, 2026

Dear Prof,

I am Adarsh, an M.Sc. student in Data Science (tentative finish date in May 2026) with a background in Applied Mathematics and Statistics (B.Sc. in Math & Stats). Previously I was a ML Research Intern at Indian Institute of Information Technology (IIIT) Kottayam from May 2025 - September 2025, where my responsibilities included developing vision-language models (VLMs) for the i2CS research group, by developing Vision Language Model architectures for Plant Disease Diagnosis, gaining experience in building and evaluating multimodal systems.

My research investigates how neural representations align with latent spaces of modern generative models, with a focus on identifying where linear mappings fail and what this reveals about representation mismatch in brain-to-image reconstruction and clinical AI systems. This work lies at the intersection of NeuroAI and generative modeling, where I aim to understand what the limitations of these models reveal about representation learning. Earlier, I have demonstrated strong hands-on proficiency in the deep learning stack. As outlined in the project description and to align with my pertinent experience, the following project summaries illustrate the diversity of my completed and ongoing research projects.

My MSc thesis examines how representations align across neural systems and generative models through two complementary studies.

(1) Brain2VLM: Hierarchical Alignment Between Cortical Representations and Vision-Language Latent Spaces. This work investigates how neural activity aligns with latent spaces of diffusion-based vision-language models for brain-to-image reconstruction. While prior work demonstrates strong reconstruction performance using latent diffusion models (1; 2), the structure of the brain-to-latent mapping remains poorly understood. I evaluate whether linear mappings are sufficient for decoding different types of representations, using ridge regression and a nonlinear residual MLP on fMRI data from the Natural Scenes Dataset (3). Prior work suggests hierarchical organization of visual and semantic representations (4; 5), motivating this analysis. I show that linear mappings are sufficient for structural diffusion latents but fail for semantic embeddings, where nonlinear decoding yields substantial improvements ($\Delta corr \approx 0.47$ vs. 0.05–0.06). However, increased decoder expressivity introduces distribution shifts that degrade generative compatibility, revealing a trade-off between alignment accuracy and reconstruction fidelity. Despite its simplicity, this approach achieves competitive reconstruction performance ($PixCorr \approx 0.33$, $CLIP \approx 85\%$) relative to recent diffusion-based approaches such as Brain-Diffuser and MindLDM (2; 6). This suggests that the brain-to-latent interface is a primary bottleneck in current decoding systems, highlighting a fundamental mismatch between neural representations and generative model latent spaces. A glimpse of the results (Brain2VLM) is shown Figure 1 below.

(2) SkinGenBench: Generative Model and Preprocessing Effects for Synthetic Dermoscopic Augmentation in Melanoma Diagnosis. (7). This work addresses the lack of reproducible, clinically grounded benchmarks for synthetic dermoscopic data (8; 9), particularly in underrepresented melanoma cases, which remain scarce in public datasets such as HAM10000 (10) and ISIC (8). I evaluate how generative model choice (GANs vs. diffusion

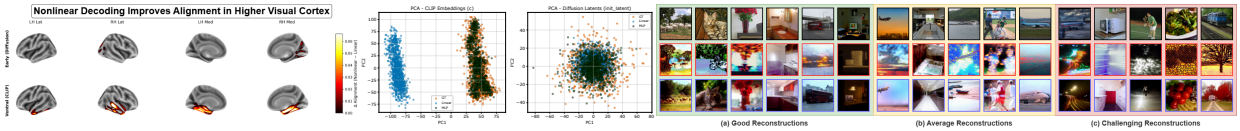

Figure 1: **in the row, Cortical surface based brain maps of alignment differences; PCA of ground-truth (GT), linearly decoded and MLP decoded CLIP latents; GT (top), linear decoder (middle), and MLP (bottom) based samples of good, average, and challenging reconstructions.**

models) and preprocessing complexity affect both image quality and downstream diagnostic performance, comparing StyleGAN2-ADA and diffusion models (11). I find that StyleGAN2-ADA consistently outperforms diffusion models in fidelity ($FID \approx 65.5$, $KID \approx 0.05$) and diagnostic utility, improving melanoma F1 by 8–15%, while diffusion models exhibit higher variance but reduced perceptual coherence, consistent with recent comparative findings (12). This challenges the assumption that diffusion models are universally superior, showing that in low-data clinical settings, GAN-based approaches remain more reliable for diagnostic tasks. A glimpse of the results (SkinGenBench) is shown in Figure 2 below.

Figure 2: **In the row, t-SNE embeddings of synthesized mel samples, FID, IS, and KID scores (epochwise, throughout training), GradCAM-visualisation (Top 2 best performing classifiers), Sample input and output of the artifact removal algorithm (advanced preprocessing)**

My work is driven by the following research questions: (1) How do neural representations align with latent spaces of generative models, and where do linear mappings fail? (2) How does decoder expressivity influence alignment and generative compatibility in brain-to-image reconstruction? (3) How does generative model choice affect diagnostic performance in low-data clinical settings?

I have delivered invited seminars on ML research methodology, covering experimental design, reproducibility, and practical workflows. I have implemented core deep learning components from first principles, including a transformer-based language model (13), with technical blogs on tokenization and attention (14; 15). My applied work includes an end-to-end life expectancy modeling pipeline using WHO data, including multicollinearity analysis (16), and a sign language translation system integrating computer vision and NLP, presented at an IEEE conference (17).

With growing interest in extending this to multimodal neural data, I see this opportunity as a continuation of my work in NeuroAI, particularly in understanding how brain signals align with latent representations of modern generative and vision–language models. I bring experience in building end-to-end deep learning pipelines across generative modeling, large-scale data, and representation learning, with a strong emphasis on reproducibility and experimental rigor. I am comfortable working in interdisciplinary settings and am keen to contribute to research at the intersection of machine learning, neuroscience, and cognitive science.

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